UNIT 1 · TOPIC 4(A)

Fertilizer Application Equipment

Syllabus topic: Equipment for fertilizer application and inter-cultivation

Broadcasters

- **Centrifugal (spinning-disc) spreader:** swath **6–12 m**. Its distribution pattern is **triangular**, so overlap the passes.
- **Pendulum spreader** (oscillating spout).
- **Aircraft.**

Band placement

- Fertilizer attachments on seed drills and planters meter granular fertilizer with star-wheel, fluted-roller, auger or agitator feeds (Kepner ch. 13).
- A seed-cum-fertilizer drill places fertilizer 5 cm below and 5 cm beside the seed (see Topic 3c).

Why fertilizer is hard to meter

Fertilizers are **hygroscopic and corrosive**. Fit hoppers with agitators and use corrosion-resistant parts (stainless steel/plastic).

Other applicators

- **Deep-placement applicators** for urea briquettes/USG (urea super granules) in paddy.
- **Liquid fertilizer / anhydrous ammonia applicators** inject behind knife openers.
- **Fertigation** delivers fertilizer through drip.
- **Manure/FYM spreaders:** a box with an apron conveyor and beaters.

Practice MCQs

Q33. Fertilizer hoppers require agitators and corrosion-resistant materials because fertilizers are:

(a) Volatile
(b) Hygroscopic and corrosive
(c) Explosive
(d) Oily

Q34. Spinning-disc broadcaster distributes fertilizer in a pattern best described as:

(a) Uniform rectangular
(b) Triangular, needing overlap
(c) Ring shaped
(d) Random

Answers

Q33 — (b) Hygroscopic and corrosive
Q34 — (b) Triangular, needing overlap